Machine Learning Fundamentals

Tags: ml, python, ai, optimization | Last Updated: Nov 05, 2026 22:56 UTC | DevNot

Gradient descent is an optimization algorithm used to minimize the loss function.

CODE (PYTHON)

```python
learning_rate = 0.01
gradient = 3.5
weight = 10.0
new_weight = weight - learning_rate * gradient
print(f'Updated Weight: {new_weight:.4f}')
```

OUTPUT (20.72ms)

```
Updated Weight: 9.9650
```

Notice how the parameter step reduces the loss proportionally to the gradient.

Python Decorators

A decorator is a callable that takes another function as an argument, extends its beh
avior without modifying it, and returns the modified function.

CODE (PYTHON)

```python
import functools
import time

def timer(func):
    @functools.wraps(func)
    def wrapper(*args, **kwargs):
        start = time.perf_counter()
        val = func(*args, **kwargs)
        print(f'{func.__name__} executed in {time.perf_counter() - start:.4f}s')
        return val
    return wrapper

@timer
def compute():
    return sum(i * i for i in range(100_000))

print('Result:', compute())
```

HANDWRITTEN NOTE: Loss Curve Graph

Continued editing after round-trip PDF import!